WHERE THE LOOPS COME FROM

Why some networks branch once, while others build a way around

THE ARGUMENT IN ONE MINUTE

FROST ON A WINDOW grows outward and never comes back. Each new arm reaches into air still carrying water vapour, splits where the supply becomes uneven, and closes nothing behind it. A **dendrite** looks like a tree, but it does not function as a transport network. It is a record of where crystal growth was easiest.

A river system is usually a tree too, for a harder reason: gravity fixes the overall direction, the main outlet does not move, and a network carrying water downhill gains little from an extra route unless flows divide and rejoin in braids, anabranches or deltas.

Put a slime mould on an agar plate with food at scattered points and it does something a river cannot. It floods the whole dish first, then thins the tubes that carry little and thickens the ones that carry much. What remains is mostly economical and direct, yet it keeps a few redundant links that no shortest-path calculation would have paid for. Break one and traffic reroutes.

Fungal networks under a forest floor can reorganise across larger distances and longer times, and cities do it across centuries. In a city no single person solves the whole problem; planners, builders and travellers each solve only a piece.

Four systems, one question: when is a network allowed to close a loop? Much of the answer cuts across chemistry, biology and geography. It depends on whether the pattern of demand stays fixed, and whether routes must survive damage.

Growth without transport

A crystal is the limiting case, and it is worth starting there because it is the one system on this list that is not solving a transport problem. Snowflake arms and the dendrites in a cooling alloy branch for a local reason: a tip that projects slightly ahead of its neighbours has better access to what growth requires - water vapour, atoms moving through a melt, or a route for heat to escape. It therefore grows faster, while the hollows between arms are starved.

This is the **Mullins-Sekerka instability**. The resulting shape is tree-like and usually does not close, but it is not a network carrying traffic. It is a frozen record of a **diffusion field**. There is no useful or redundant branch because no branch has a transport job.

One important control on the final shape is the **bottleneck**: whether atoms attach slowly at the surface or whether heat and material diffuse slowly through the surroundings. Temperature, concentration and the preferred growth directions of the crystal matter too. Even here, shape records the process that limited growth.

Hold onto the word *bottleneck*, because the rest of the argument is a sequence of systems in which the bottleneck stops being growth and starts being delivery.

Trees when demand is fixed

Most river drainage networks branch downstream without rejoining. Braided reaches, anabranching channels and deltas are important exceptions. Away from them, optimal channel network theory models the drainage basin as a system that reduces the total energy spent moving water downhill. The near-tree structure is encouraged by one overall direction of flow and a stable outlet.

The geometry is related to the **Steiner tree problem**: connect a fixed set of points using the least total length. A minimum-length connected network contains no **cycle**, because an edge can be removed from any cycle while leaving all the points connected. The network then has less length, so the cycle could not have been part of the minimum.

The first rule is therefore more precise than “certainty produces trees”. Fixed demand makes loops hard to justify. One fixed source and one fixed sink require only a path; several fixed terminals can produce a tree. When sources, sinks or failures vary, minimum length is no longer the whole objective.

Loops when demand moves

Physarum polycephalum, a single-celled slime mould containing many nuclei, faces a problem no ordinary drainage tree faces. Food lies at unpredictable points, the organism begins without a map of the useful routes, and any tube can be severed. Its rule is startlingly simple: spread widely, then thicken tubes carrying high **flux** and let low-flux tubes shrink.

Tero and colleagues famously placed oat flakes at locations representing towns around Tokyo. The resulting slime-mould network was comparable with the rail system in total length, transport efficiency and **fault tolerance**. That statement needs care: the mould did not design a railway. Researchers measured a similarity among three network properties.

What the mould produces is not quite a tree. It keeps some loops. Work by Corson, Katifori and colleagues on leaf veins showed why: when a transport network is optimised for fluctuating loads or possible damage, cycles can enter the best solution. Fixed sources and sinks under steady flow favour a tree; moving demand or broken links can make redundancy worth its cost. Leaves do this, as do many vascular networks in organs and the slime mould itself.

That is the spine of the argument. Loops are what can become worth paying for when a network does not know where demand will be or which route will survive.

Underground, with a necessary caution

Mycorrhizal fungi can connect plant roots, while **mycelial networks** foraging through soil reorganise by thickening productive cords and abandoning others. The same trade-off may apply: unpredictable resources, grazing and disturbance can reward redundant routes. The evidence that a whole forest network is optimising in exactly the same way as a slime mould is much less direct.

The popular account goes considerably further, into a "wood-wide web" in which mother trees deliberately feed seedlings through fungal cables and warn neighbours of insect attack. Karst, Jones and Hoeksema challenged that framing after finding that influential claims about **common mycorrhizal networks** were weakly supported and repeatedly amplified by later citations.

It is worth being precise about what survives that criticism. Fungal hyphae form networks and can connect roots; carbon compounds move among organisms and soil. What remains contested is how much moves directly between trees through shared fungal pathways, in which direction, who benefits, and above all whether intent should be inferred. The fungus is not simply a public utility. The loops can first be understood as supporting fungal foraging and resilience; no forest-wide purpose is required.

Cities, where nobody has the whole problem

Most large street systems contain many loops, although cul-de-sac neighbourhoods are deliberately tree-like. City destinations change hourly, everyone can be both source and sink, and any route can be blocked. Cities differ in who does the adapting: there are planners, but no single planner solves the whole network across centuries.

A city's streets are the accumulated result of planned and unplanned decisions, constrained by a river crossing here and an old field boundary there. Grids, radials and medieval tangles can therefore sit beside one another inside one boundary, and a fast route across the modern city is often the fossil of an older purpose.

There are statistical regularities amid that history. Bettencourt, West and colleagues reported that several measures of infrastructure grew **sublinearly** with population, while several measures of economic and social output grew **superlinearly**. Other analyses have questioned how universal or predictive those power laws are. The safe conclusion is that strong patterns can emerge across cities without every street sharing one design, while the exact scaling claim remains an empirical question.

Regularity without a single planner is what one might expect when many local decisions meet the same transport constraints.

The decisive variable

The four systems arrange themselves by two questions, and the disciplines fall out afterwards.

The first is whether the structure functions as a transport network. A crystal does not, so its branching answers to the diffusion field around it. The others carry something, so they face costs and competing objectives.

The second is whether the load and route remain fixed. Where demand is steady, routes are reliable and length dominates the cost, a tree is favoured. Where loads fluctuate or links can fail, redundancy may pay for itself and cycles can appear.

What differs among the last three cases is the mechanism and timescale of adaptation: minutes to hours and centimetres of flux-driven tube reinforcement in the mould; seasons and metres to hectares of growth and abandonment in fungi; decades to centuries and square kilometres of human decision in a city. A related compromise is reached by a single cell with no nervous system, a branching organism with no fixed body plan, and a population of people none of whom is solving the whole problem.

The one sentence

A loop is a confession that the future is unknown.

More exactly, it is what can become worth paying for when demand moves or routes fail. That is why loops are common in adaptive transport networks and rare in ordinary drainage trees. Braids, anabranches and deltas are the exceptions that keep the aphorism honest.

SELECTED EVIDENCE

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